

Create R script for visualization

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```
# Step 1: Load required packages
```

```
library(ggplot2)
```

```
# Step 2: Import synthetic dataset
```

```
data <- read.csv("practical-application/synthetic_data.csv")
```

```
# Step 3: Create histogram of PageViews by Version
```

```
pageviews_plot <- ggplot(data, aes(x = PageViews, fill = Version)) +  
  geom_histogram(position = "dodge", bins = 15) +  
  theme_minimal()
```

```
# Step 4: Save PageViews histogram
```

```
ggsave(  
  filename = "outputs-figures/pageviews_hist.png",  
  plot = pageviews_plot,  
  width = 6,  
  height = 4  
)
```

```
# Step 5: Create histogram of TimeSpent by Version
```

```
timespent_plot <- ggplot(data, aes(x = TimeSpent, fill = Version)) +  
  geom_histogram(position = "dodge", bins = 15) +  
  theme_minimal()
```

```
# Step 6: Save TimeSpent histogram
```

```
ggsave(  
  filename = "outputs-figures/timespent_hist.png",  
  plot = timespent_plot,  
  width = 6,  
  height = 4  
)
```